

Internship Report

on

Study of Industrial Automation, Control Systems, and Data-Driven Monitoring in Utility Plants at Adani Kutch Copper Limited

Including a Proof-of-Concept Data Analysis for Predictive
Monitoring of a Compressed Air System

Submitted By:

Name: Vedant Tejas Vora

SRN: PES1UG24EC307

Class: ECE 'F' Section

Internship Organization:

Adani Kutch Copper Limited, Mundra

Department:

Control and Instrumentation

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I. Introduction

This report presents the internship undertaken in the Control and Instrumentation department at Adani Kutch Copper Limited (KCL), Mundra. KCL is a large-scale copper refinery that processes imported copper concentrates into refined copper and several valuable by-products. The operation of such a plant requires complex process systems, advanced automation, and precise control mechanisms to ensure efficiency, safety, and continuous production.

The copper refining process begins with imported raw material, which is first handled through laboratory testing and warehouse storage. It then passes through major processing stages such as flash smelting, Pierce-Smith conversion, anode furnace operation, anode casting, electro-refining, and electro-winning to obtain high-purity copper. Additional units such as nickel recovery and effluent treatment support resource recovery and environmental compliance. Refined copper is further directed to the Continuous Casting and Rolling (CCR) unit for producing finished copper products, while by-products are recovered at different stages of the process.

Along with the main copper production units, the plant is supported by several critical utility systems. The sulphuric acid plant converts sulphur dioxide generated during smelting into concentrated sulphuric acid, reducing emissions and producing a useful by-product. Water treatment systems supply potable water for general use and demineralized water for cooling and industrial applications.

One of the key utility systems is the Air Separation Unit (ASU), which produces oxygen, nitrogen, and argon through cryogenic distillation. Oxygen is supplied to the smelting process, while liquid nitrogen and argon are stored and sold as by-products. The plant also has a centralized compressed air system, which supplies instrument air and utility air across different sections of the plant. These utilities are closely integrated with the main process and depend heavily on instrumentation, automation, and control systems.

The Control and Instrumentation department plays an important role in maintaining the stability and reliability of both process and utility systems. The department is involved in field instruments such as pressure, temperature, flow, and level sensors, along with their integration into monitoring systems

such as the Distributed Control System (DCS). These systems enable real-time data acquisition, process visualization, alarm monitoring, and closed-loop control.

From an Electronics and Communication Engineering perspective, the plant provides practical exposure to concepts such as signal transmission, data acquisition, industrial communication, and control systems. The interaction between sensors, transmitters, controllers, communication networks, and actuators forms the backbone of industrial automation, allowing process parameters to be monitored and regulated accurately.

During the internship, focus was placed on understanding the instrumentation, control logic, and operational behavior of key utility systems, particularly the Air Separation Unit (ASU), Water Treatment Plant (WTP), and Compressed Air System. Special emphasis was given to studying equipment functions, control loops, signal flow, and data exchange between field instruments and the DCS. The ASU was studied in greater detail due to its complexity and its critical role in supplying oxygen for smelting operations.

This report combines process understanding with practical observations from the control room and field visits. It includes an overview of the utility systems studied, industrial automation architecture, data acquisition methods, instrumentation systems, PID control loops, and real-time plant behavior. The objective is to highlight the relevance of Electronics and Communication Engineering in modern industrial automation and document the technical learning gained during the internship.

2. Overview of Utility Systems Studied

During the internship, exposure was primarily focused on the utility systems that support the operation of the copper refinery. These systems provide essential resources such as process water, demineralized water, oxygen, nitrogen, argon, and compressed air, which are required for the continuous and efficient functioning of various plant units. The utility systems studied in detail were the Water Treatment Plant (WTP) and the Air Separation Unit (ASU), both of which rely heavily on instrumentation, automation, and process control.

2.1 Water Treatment Plant (WTP)

The Water Treatment Plant receives treated desalinated water from a reservoir supplied by Adani Power Limited. The water is initially pumped through a multimedia filter to remove suspended impurities, followed by an ultrafiltration (UF) basket and UF skid for finer filtration. The filtered water is then collected in a UF permeate tank, from which the process splits into two separate streams.

The first stream is the potable water system. Water from the UF permeate tank passes through a limestone filter and is stored in a potable water tank. It is then dosed with sodium hypochlorite (NaOCl) for disinfection before being stored in a hydropneumatic tank and distributed through the potable water network for domestic applications such as canteens and other facilities.

The second stream is the demineralized (DM) water system. Water is chemically conditioned using acid, anti-scalant, and sodium metabisulfite (SMBS) before entering the reverse osmosis (RO) system. Following storage in an RO feed tank, the water passes through a cartridge filter, high-pressure pump, and a two-stage RO skid. The permeate then enters a degasser tower where dissolved carbon dioxide is removed. After collection in the RO permeate tank, the water is pumped through a mixed-bed unit for final polishing before being stored in the DM water tank. This high-purity water is used for cooling and other critical industrial applications within the plant.

2.2 Air Separation Unit (ASU)

The Air Separation Unit is responsible for producing oxygen, nitrogen, and argon through a cryogenic separation process. Atmospheric air is drawn into the system through a feed air filter and compressed using a multi-stage Main Air Compressor (MAC), where the air undergoes alternate stages of compression and cooling. The compressed air is then sent to a pre-cooling system to further reduce its temperature.

Following pre-cooling, the air passes through a molecular sieve purification unit containing molecular sieve and activated alumina. This stage removes moisture, carbon dioxide, and hydrocarbons that could otherwise freeze and disrupt cryogenic operation. The purified air is then routed through the main heat exchanger and distributed to different sections of the process, including the expansion turbine, booster air compressor, and distillation system.

Within the cryogenic distillation columns, air is separated into its constituent gases based on their boiling points. Liquid nitrogen from the upper column is stored and supplied to users, while oxygen-rich liquid from the lower column undergoes further processing to produce liquid oxygen and high-purity oxygen products. A portion of the argon-rich stream is directed to crude and pure argon columns, where argon is separated and stored as liquid argon. The process incorporates multiple internal recycle streams and heat integration mechanisms to improve separation efficiency and reduce energy consumption.

2.3 Compressed Air System (CAS)

The Compressed Air System is one of the important utility systems in the plant, supplying compressed air for various operational and control-related requirements. The system consists of four compressors, which draw atmospheric air through feed filters before compression. Each compressor carries out compression in three stages, with cooling provided between stages to control the rise in air temperature. The compressed air is water-cooled using cooling water supplied from the cooling towers, where the water is circulated and reused for heat removal.

Each compressor also has an associated oil system for lubrication. Oil from the reservoir is supplied to the compressor for lubrication of moving parts and is then passed through oil coolers to remove heat before being recirculated. This helps maintain the reliability and safe operation of the compressor units during continuous service.

The compressed air produced is distributed across different sections of the plant, including the Air Separation Unit, smelter area, and other utility or process areas. A major application of compressed air is in the operation of pneumatic control valves, where instrument air is used to open, close, or modulate valve positions based on control system commands.

This overview provides the process context for the subsequent sections, which focus on the instrumentation, control systems, data acquisition architecture, and PID-based automation observed within these utility plants. From an instrumentation and control perspective, both WTP and ASU contain multiple process variables that require continuous monitoring and regulation. Parameters such as pressure, flow, level, temperature, and conductivity are measured through field instruments and monitored through centralized control systems. Therefore, these utility systems provided a suitable environment to study industrial automation, signal transmission, and closed-loop control in practical operation.

3. Industrial Automation and Communication Systems

3.1 Role of Automation in Large-Scale Industrial Plants

During the internship, it became clear that large-scale industrial plants depend heavily on automation systems to continuously monitor, control, and coordinate multiple interconnected processes from centralized control rooms. To achieve this, industrial automation systems are employed to ensure stable operation, minimize human intervention, and improve process efficiency and safety. In process industries such as copper refining and utility generation, automation enables operators to supervise large sections of the plant from centralized control rooms while maintaining precise control over critical parameters such as pressure, flow, temperature, and level.

3.2 SCADA and DCS Interface

During the internship, exposure to the Distributed Control System (DCS) and SCADA interfaces provided practical insight into the implementation of industrial automation in a real operating environment. The graphical displays available within the control room presented a comprehensive view of plant operations, including equipment status, instrument readings, valve positions, alarms, process trends, and active control loops. Through these interfaces, it was possible to navigate across different sections of the utility plants and observe how process variables were monitored and controlled in real time.

The DCS architecture serves as the central layer of the automation system, collecting data from field instruments distributed throughout the plant and presenting it to operators in an organized manner. This architecture enables the integration of thousands of process signals while simultaneously executing control logic and alarm management functions. A key observation during the internship was that operators rarely interact directly with field devices; instead, all monitoring and control activities are performed through the DCS interface, which acts as the link between the physical process and the control room.

3.3 Industrial Communication Network

Communication between various automation components is achieved through industrial networking technologies. Information from field-level devices is transmitted through Remote I/O stations and communicated across the control network using industrial Ethernet-based protocols such as PROFINET. These communication systems provide reliable and high-speed transfer of process information while supporting continuous plant operation. The distributed nature of the architecture allows instrumentation installed across large geographical areas to be integrated into a single centralized control platform.

3.4 Troubleshooting and Commissioning Observations

One of the most valuable learning experiences was observing how the automation infrastructure supports troubleshooting and commissioning activities. During the commissioning phase of a nitrogen blower system, valve operations were tested directly from the control room by issuing commands through the DCS interface and verifying the response both on-screen and in the field. This demonstrated how automation systems are used not only for routine plant operation but also for equipment validation and fault identification. It highlighted the close relationship between control logic, communication systems, and field instrumentation in ensuring reliable plant performance.

3.5 Relevance to Electronics and Communication Engineering

The study of the DCS and communication architecture provided a practical understanding of how Electronics and Communication Engineering principles are applied in industry. Concepts such as distributed control, industrial networking, centralized monitoring, and real-time communication were observed in operation, providing a strong foundation for understanding the data acquisition and signal processing systems discussed in the following section.

4. Data Acquisition and Signal Transmission Systems

4.1 Understanding the Signal Path from Field to DCS

After understanding the overall automation and communication architecture of the plant, the next focus was to understand how field data is acquired and made available for monitoring and control. In the utility plants, field instruments measure physical parameters such as pressure, flow, temperature, and level. These measured values are not directly used by the operator; they first pass through a signal conversion and transmission path before appearing on the DCS/SCADA interface.

During the internship, this signal path was understood by observing field instruments, Remote I/O panels, DCS graphics, live trends, and control loop displays. This helped in relating theoretical concepts of Electronics and Communication Engineering to actual industrial applications, especially in the areas of analog signals, digital conversion, communication networks, and control system interfacing.

4.2 Field Instruments and 4–20 mA Signal Transmission

Most industrial instruments observed in the plant, such as pressure and flow transmitters, convert the measured process variable into a standard electrical signal. A commonly used signal range is 4–20 mA, where 4 mA represents the lower range of measurement and 20 mA represents the upper range. This standard is widely used in industrial environments because it is reliable for long-distance transmission and less affected by electrical noise compared to voltage signals.

For example, if a pressure transmitter is designed for a range of 0–10 bar, then 4 mA would correspond to 0 bar and 20 mA would correspond to 10 bar. Any intermediate pressure value is represented by a proportional current signal. Understanding this conversion was important because the values displayed on the DCS screen are ultimately derived from these field-level electrical signals.

4.3 Signal Conditioning and Data Reliability

One important observation during control room monitoring was that the values displayed on the SCADA system were generally stable and responsive to changes in process conditions. This indicated that the instruments, communication system, and signal conditioning elements were functioning reliably. In a large industrial plant, maintaining signal quality is important because incorrect or noisy readings can affect monitoring, alarms, and control decisions.

Proper instrumentation design, shielding, grounding, and signal conditioning help ensure that the signal received by the control system is clean and accurate. This is especially important in industrial environments where electrical noise, long cable distances, and heavy machinery can affect signal quality.

4.4 Analog-to-Digital Conversion at Remote I/O Panels

The analog signals generated by field instruments are received at Remote I/O panels. At this stage, Analog-to-Digital Conversion (ADC) takes place. The ADC converts the continuous 4–20 mA analog signal into digital data that can be processed by the control system.

This conversion is essential because the DCS and controllers operate digitally. The analog field signal must therefore be converted into digital form before it can be used for calculation, display, alarm generation, or PID control. Through this process, physical measurements from the plant are transformed into usable digital information for monitoring and automation.

4.5 Fiber Optic Communication for Long-Distance Transmission

Another important learning was the use of optical communication for long-distance signal transmission. Since the plant covers a large area, transmitting electrical signals directly over long distances can lead to losses, noise pickup, and interference. To overcome this, some electrical data signals can be converted into optical signals and transmitted through fiber optic cables.

Fiber optic communication helps reduce electromagnetic interference and allows reliable high-speed data transmission between field areas and the control room. This is especially useful in industrial environments where heavy motors, compressors, and electrical equipment operate continuously. The use of optical transmission also supports centralized monitoring by allowing data from distant plant sections to reach the DCS with minimum degradation.

4.6 Digital Processing and Output Signal to Actuators

Once the digital data reaches the DCS, it is processed and displayed on the SCADA interface in the form of numerical values, trends, alarms, and control loop graphics. If the parameter is part of a control loop, the measured process variable is compared with the setpoint. Based on the error between the measured value and the desired value, the controller generates a corrective output.

When a control action is required, the digital output signal from the controller is converted back into an analog signal through Digital-to-Analog Conversion (DAC). This output is commonly sent as a 4–20 mA signal to final control elements such as control valves. Based on the received signal, the valve changes its opening percentage and regulates the process variable.

4.7 Process Data Logging and Analysis Scope

In addition to observing live values, a structured effort was made to log and organize process data from the utility systems. Around fifty PID control loops were recorded from the Air Separation Unit, including process variables, manipulated variables, equipment tags, and remarks wherever possible. Major machinery tags from the ASU were also documented, along with alarm setpoints from the Water Treatment Plant and instrument details from the Compressed Air System.

This activity helped in understanding how different process variables are connected to control actions. It also showed that plant data is not just a set of fixed values but a time-dependent signal that changes with operating conditions. Trends, fluctuations, alarms, and responses to valve movements can all provide useful information about system behavior.

The collected information also provides a base for further analysis using signal processing and data analytics techniques. Once time-series data is available, it can be used for plotting trends, converting engineering values into 4–20 mA signals, applying filters, studying signal fluctuations, and identifying abnormal behavior in plant operation.

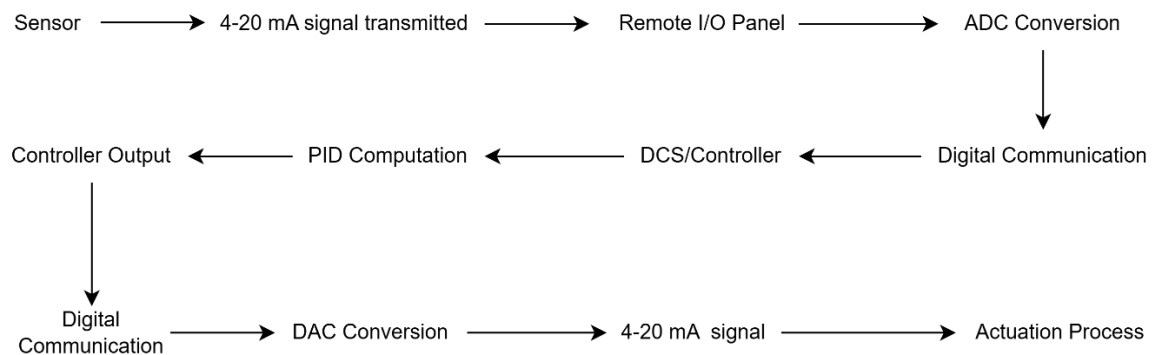


Figure 1: Complete Information Flow Graph

4.8 Summary

This section helped connect the field instrumentation layer with the digital control system used in the plant. The study of signal transmission, ADC/DAC conversion, fiber optic communication, and data logging provided a practical understanding of how Electronics and Communication Engineering concepts are applied in industrial automation.

5. Control Systems

5.1 Role of Closed-Loop Control in Plant Operation

The effective operation of industrial processes at the plant is heavily dependent on the implementation of closed-loop control systems, particularly using PID (Proportional–Integral–Derivative) controllers. During the internship, it was observed that multiple control loops operate simultaneously across different subsystems, regulating key parameters such as pressure, flow, level, and temperature.

Each control loop follows a structured methodology, where a process variable (PV), measured through field instrumentation, is continuously compared with a desired setpoint (SP). Based on this comparison, the controller computes a corrective output, referred to as the manipulated variable (MV), which is transmitted to final control elements such as valves or compressor control systems to maintain process stability.

Through interaction with the DCS interface, it was possible to observe around fifty such PID loops within the Air Separation Unit (ASU), each governing a specific aspect of the process. These loops work in coordination to ensure stable operation, particularly in critical sections such as air compression, cooling, and cryogenic separation, where even small deviations in parameters can impact overall system performance. The ability of PID controllers to respond to dynamic process variations in real time plays a crucial role in maintaining efficiency, safety, and continuity of plant operations.

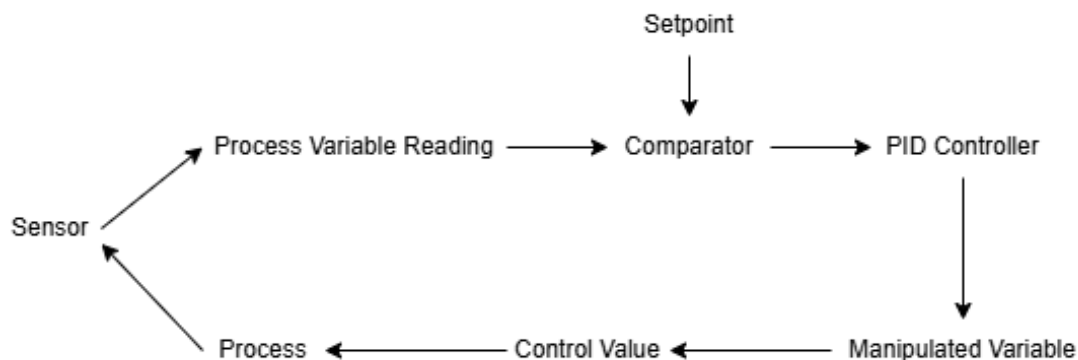


Figure 2: General Control Loop

5.2 Advanced Control Strategies Observed in ASU

Beyond conventional single-loop control, several advanced control strategies were observed within the Air Separation Unit, particularly in sections involving dynamic process interactions. One notable example was the implementation of cascade control logic in the booster air compression system, where a master pressure control loop, PIC220115_1, governs the desired pressure, and its output is used to adjust a secondary loop associated with the actuator, UIC220111, which controls the valve position in terms of percentage opening.

This hierarchical control structure improves system responsiveness by allowing the inner loop to react faster to disturbances, thereby stabilizing the overall process more effectively. A similar cascading behavior was observed in the expansion turbine section, where turbine performance parameters such as rotational speed (RPM) and flow control were interlinked, enabling coordinated regulation of energy extraction and temperature reduction within the cryogenic cycle. This unique control architecture is seen in the diagram below, where the master controller SIC040101A produces the output signal which becomes the setpoint for the secondary controller UIC040101A. This controller achieves this set point by changing the manipulated variable, which is the valve position of the valve FV040104A.

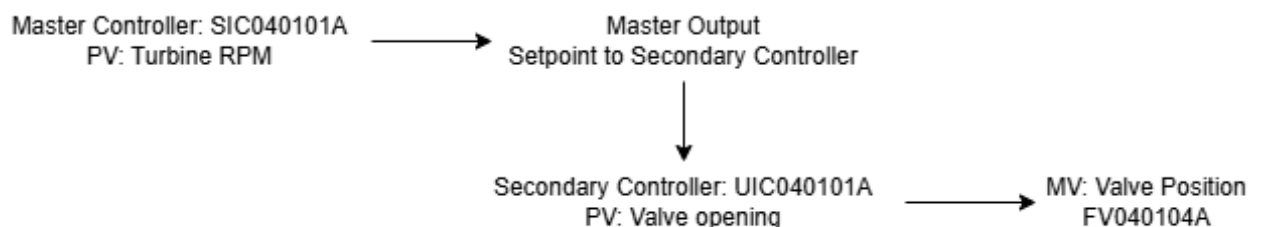


Figure 3: Cascaded Control loop Architecture

5.3 Differential Pressure Control in Distillation Section

In addition to cascade control, differential pressure control was also observed in the distillation column, where maintaining a precise pressure difference between the upper and lower sections of the column is critical for effective separation. This was achieved through a dedicated control loop, PdIC054707 which continuously monitors the pressure difference and adjusts a corresponding control valve to maintain the required gradient.

Unlike standard pressure control, this loop does not regulate an absolute value but instead ensures a stable relationship between two process variables. This highlights the complexity of control strategies employed in separation processes. These examples demonstrate that industrial control systems extend beyond basic PID loops and involve coordinated, multi-layered control architectures designed to handle the highly sensitive and interdependent nature of processes within systems such as the ASU.

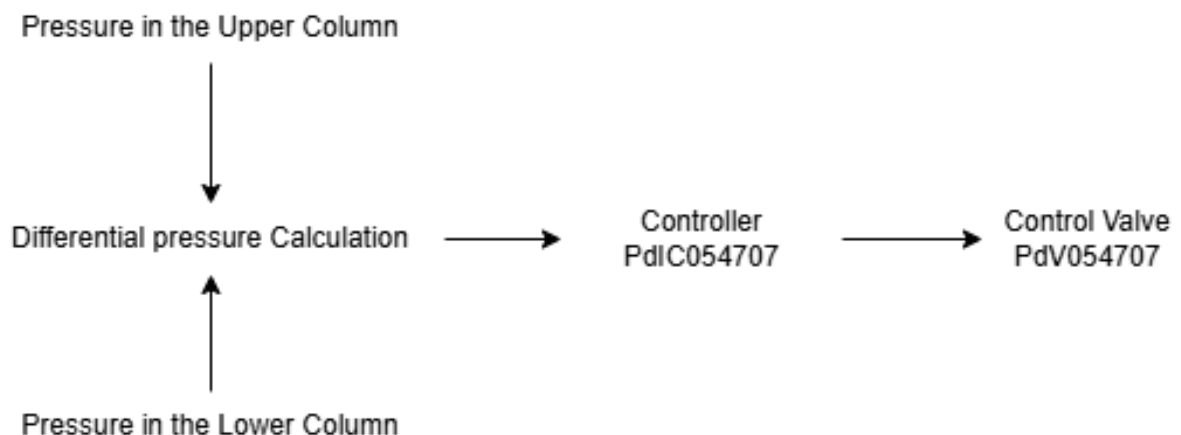


Figure 4: Case study of the Differential Pressure PID Control Loop

5.4 Troubleshooting and Control Tuning Observations

During the observation period in the Air Separation Unit, the practical aspects of control system tuning and troubleshooting were clearly demonstrated by a commissioning team working on the system. In particular, attention was focused on optimizing argon production, which was not performing as expected and was identified as a critical issue due to its economic impact.

The troubleshooting approach adopted involved making very fine adjustments to control valves, often in increments as small as 0.5% opening, directly from the control system. These small perturbations were introduced deliberately to study the sensitivity of the process and observe how different variables responded. By monitoring real-time trends on the SCADA interface, the team analyzed the behavior of associated parameters such as pressure, flow, and temperature in the surrounding sections of the plant.

5.5 Importance of Trend-Based Control Analysis

This method highlighted an important aspect of industrial control systems — even minor changes in manipulated variables can lead to noticeable variations in process conditions, especially in tightly coupled and highly sensitive systems such as cryogenic separation units. The use of trend graphs allowed for visualization of dynamic responses, helping identify delays, oscillations, or stabilizing behavior in the system.

Rather than relying solely on theoretical tuning, the control strategy involved iterative adjustments combined with continuous observation, reflecting the practical limitations and non-linear characteristics of real-world processes. This experience emphasized the importance of fine control resolution, system responsiveness, and operator expertise in maintaining optimal plant performance, particularly in processes where multiple variables are interdependent and directly affect product yield.

6. Process Data, Trend Monitoring and Historian Systems

6.1 Role of Process Data in Plant Operation

After studying the automation architecture, data acquisition system, and control loops, it became important to understand how process data is stored, visualized, and used by operators. In industrial systems, process data is not limited to live values on the DCS screen; it is also used for trend monitoring, alarm review, troubleshooting, and operational decision-making.

During the internship, different approaches to data visualization and storage were observed across the Water Treatment Plant, Air Separation Unit, and Compressed Air System. Each system had a different level of data availability depending on its DCS architecture, commissioning status, and integration with centralized data systems.

6.2 Trend and Alarm Monitoring in WTP

At the Water Treatment Plant, the DCS was observed to be a localized system with trend monitoring available for several instruments. Trends were available for parameters such as pressure, flow, differential pressure, level, and other process variables. These trends allowed operators to observe how process values changed over time instead of relying only on instantaneous readings.

The WTP system also had alarm logs where past alarms were visible for review. These logs helped in understanding how the DCS supports process safety, abnormal-condition monitoring, and setpoint-based operation. However, the trend and alarm data appeared to be primarily available for operator viewing through the DCS interface, with limited direct exportability for external analysis.

6.3 Trend-Based Troubleshooting in ASU

The Air Separation Unit also operated on a localized control system where trend graphs were used actively for troubleshooting. Some trends showed oscillating behavior, similar to the transient response seen in unstable or marginally stable control systems. This demonstrated how trend monitoring can support control tuning, process stabilization, and troubleshooting.

Another observation in the ASU was related to alarm management. Operators and engineers discussed converting certain non-essential alarms into silent alarms so that critical events could receive more attention. This highlighted the importance of alarm prioritization in industrial control systems.

6.4 Historian Data in the Compressed Air System

The Compressed Air System was different from WTP and ASU because it was not yet operating as a complete DCS-based system. The system was largely automatic, while operators could mainly observe plant behavior from the interface. Some values were manually operated, while other functions were handled through predefined automation logic.

Unlike the localized DCS systems in WTP and ASU, the Compressed Air System was connected to a centralized historian system. However, the available historian data appeared to be mainly related to trip conditions of components such as compressors and pumps. Continuous time-series trends were not visible at the CAS interface, and detailed event storage appeared limited compared to a full DCS trend system.

This made the CAS useful from an equipment reliability perspective rather than a continuous process-trend perspective. The available data seemed more suitable for studying trip events, equipment interruptions, and failure occurrences.

6.5 Summary

The observations from WTP, ASU, and CAS showed that process data availability depends strongly on the control system architecture and storage configuration of each plant. WTP and ASU provided useful trend and alarm information through localized DCS systems, while CAS provided centralized trip-related historian data. These observations helped establish the importance of structured and exportable time-series data, which is explored further in the proof-of-concept data analysis section.

7. Suggestions and Future Scope

Based on the observations made during the internship, a major future opportunity lies in improving how process data, alarm logs, trip records, and calibration history are stored and used for engineering analysis. These suggestions are not corrections to the existing system, but possible future enhancements to support better monitoring, diagnostics, predictive maintenance, and instrument reliability.

7.1 Improved Access to Time-Series Process Data

In localized DCS systems such as those observed in the Water Treatment Plant and Air Separation Unit, trend graphs were available for parameters such as pressure, flow, differential pressure, level, and temperature. However, the data appeared to be mainly available for operator viewing through the DCS interface. If technically permitted by the vendor system and plant data policy, selected time-series data could be exported for approved engineering analysis.

Such data would allow engineers to study process behavior beyond instantaneous values and alarm timestamps. Parameters such as pressure variation, flow fluctuation, valve response, compressor behavior, temperature rise, and load changes could be analyzed using filtering, trend comparison, oscillation detection, and statistical methods. For centralized systems such as the Compressed Air System, storing selected continuous trends in the historian in addition to trip-related data would help identify early signs of abnormal performance.

7.2 Structured Alarm, Event, and Trip Logging

A structured digital system for storing alarms, events, and trip records across individual plants would be useful for future analysis. Such records could include timestamps, equipment tags, instrument tags, alarm type, operating condition, operator remarks, and recovery time.

Time-series data shows how the plant behaved, while alarm and event logs explain what happened during abnormal conditions. Together, these records could support reliability analysis, maintenance planning, and future machine learning-based studies.

7.3 Machine Learning-Based Predictive Monitoring

Once sufficient time-series data, alarm logs, trip records, and maintenance history are available, machine learning-based predictive analytics can be used to estimate equipment health and identify abnormal operating patterns. Models such as Random Forests, Gradient Boosting methods, and neural networks can be trained using historical trends and event data.

Such models could help predict repeated equipment trips, detect unstable control behavior, identify abnormal process patterns, classify alarms, and support preventive maintenance decisions. For example, repeated pressure fluctuations, rising temperature, increasing motor current, or abnormal load behavior before a trip could be used to provide early warning.

7.4 Instrument Drift Monitoring and Predictive Calibration

Another possible improvement is the use of data-driven instrument drift monitoring along with scheduled calibration. Industrial instruments are generally calibrated based on fixed schedules, but some instruments may drift faster than others depending on service conditions.

If calibration records such as zero error, span error, drift direction, action taken, and calibration date are stored digitally, machine learning models can be used to identify instruments showing repeated drift patterns. This could support predictive calibration, where instruments are prioritized based on actual behavior rather than only fixed intervals.

7.5 Overall Future Scope

Overall, the future scope lies in combining time-series process data, alarm logs, trip records, and calibration history into a structured digital framework. With sufficient historical data, machine learning models can support condition monitoring, predictive maintenance, trip-risk identification, and instrument drift analysis.

For a large-scale automated plant, such systems could help improve reliability, reduce unexpected failures, support better maintenance planning, and enable more data-driven engineering decisions.

8. Proof-of-Concept Data-Driven Mini Analysis Project

8.1 Objective and Scope

This section of the report is intended to demonstrate the benefits of storing plant data in a structured and methodical manner. Properly recorded time-series data can support signal processing, data analysis, and machine learning techniques to derive insights that may not be immediately visible through routine operator observation or event logs alone. Since continuous historical sensor data from the actual plant was not available in a structured and exportable form during the internship period, a synthetic dataset was used to demonstrate the type of analysis that could be performed if such data were recorded systematically.

The purpose of this proof-of-concept was not to develop a final deployable plant model, but to show how time-series process data, alarm logs, signal processing techniques, and machine learning models could support compressor monitoring, early fault identification, and preventive maintenance planning. The analysis was focused on a compressed air system compressor and included process variables such as discharge pressure, air flow, discharge temperature, motor current, oil temperature, cooling water outlet temperature, valve opening, load percentage, vibration, running status, compressor health state, and trip-risk condition. The overall workflow was designed to demonstrate how continuous data storage can provide deeper engineering insight than event logs alone.

8.2 Data Description and Validation

The proof-of-concept analysis used a structured synthetic dataset representing the operation of a compressed air system compressor over a period of 10 days. The purpose of using this dataset was to demonstrate how continuous process data, alarm logs, trip-risk labels, and calibration records could be analyzed if similar data were stored systematically in a plant historian.

The dataset consisted of three main components: time-series compressor data, an alarm/event/trip log, and a calibration drift log. The main focus of the analysis was on the time-series data and alarm log, as these were directly

useful for compressor monitoring, signal processing, FFT analysis, and machine-learning-based trip-risk prediction.

Table 8.1: Dataset Summary

Dataset Component	Records	Validation Summary
Time-Series Compressor Data	14,400 rows	10 days of 1-minute sampled data with no missing timestamps, duplicate timestamps, or missing values
Alarm/Event/Trip Log	42 records	Alarm and trip events matched corresponding abnormal behavior in the time-series data
Calibration Drift Log	80 records	Instrument calibration and drift records were available for predictive calibration demonstration

The time-series data covered the period from 2026-07-01 00:00:00 to 2026-07-10 23:59:00, giving exactly 14,400 samples at one-minute intervals. The dataset audit confirmed that the time-series data was continuous and did not require correction for missing samples, duplicate records, or irregular timestamp intervals.

The numerical values were also checked to ensure that they follow logical compressor behavior. During running periods, pressure, flow, current, temperature, valve opening, and load values remained within realistic operating ranges. During trip or shutdown periods, pressure, flow, current, load, and running status dropped as expected.

The alarm log was cross-verified with the time-series data. Low-pressure alarms, high-temperature alarms, high-current alarms, high-vibration alarms, and compressor trip events were reflected in the corresponding process measurements at the same timestamps. Compressor trip events were also

visible in the time-series data through pressure collapse, flow reduction, motor current drop, and running status changing to zero.

Based on this validation, the dataset was considered internally consistent and suitable for proof-of-concept analysis. It was therefore used for trend analysis, alarm-context study, noise filtering, FFT-based oscillation analysis, and machine-learning-based trip-risk prediction.

8.3 Time Series Trend Analysis

Time-series trend analysis was performed to observe how the main compressor parameters changed over the 10-day operating period. The variables selected for this analysis were discharge pressure, air flow, motor current, discharge temperature, vibration, and running status. These parameters were selected because they represent compressor delivery, loading condition, thermal behavior, mechanical condition, and operating state.

8.3.1 Overall 10-Day Trend Observation

The first trend plot was generated to observe the overall compressor behavior across the complete dataset. During most of the operating period, the compressor remained in a stable running condition, with discharge pressure maintained close to the normal operating range. Flow, motor current, temperature, and vibration varied according to loading conditions.

The plot also shows distinct trip or shutdown periods where discharge pressure, air flow, motor current, vibration, and running status drop sharply. This confirms that continuous time-series data can clearly show transitions between normal operation and abnormal shutdown conditions. Such behavior would not be fully visible from alarm timestamps alone.

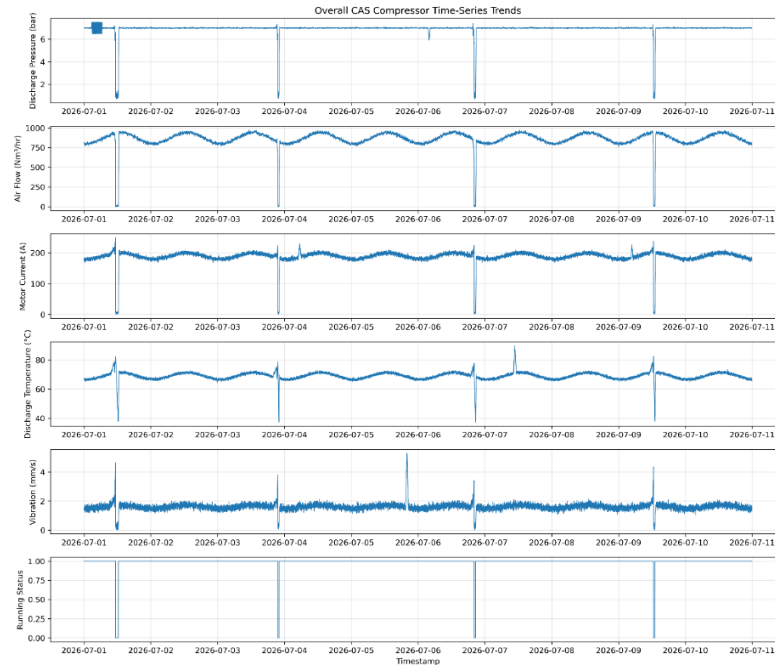


Figure 5: Overall 10-Day Trend of Key CAS Compressor Parameters

A health-state-wise comparison showed that compressor parameters changed noticeably before trip events. These changes indicate that the compressor showed measurable deterioration before entering the trip condition.

Table 8.2: Health-State Comparison of Key Compressor Parameters

Parameter	Normal Average	Trip Risk Average	Observation
Discharge Pressure	6.99 bar	6.49 bar	Pressure reduced before trip-risk condition
Motor Current	189.00 A	212.75 A	Motor loading increased before trip-risk condition
Vibration	1.60 mm/s	2.83 mm/s	Vibration increased before trip-risk condition

8.3.2 Focused July 1 Trip Case Study

A focused case study was performed around the July 1 compressor trip. This window was selected because it contains a clear sequence of abnormal behavior before the compressor trip. The zoomed trend shows that discharge pressure became unstable and then collapsed, while motor current, discharge temperature, and vibration increased before shutdown.

At the trip point, running status changed from 1 to 0, confirming that the compressor had stopped. After reset or restart, pressure and current gradually returned toward normal operating values. This focused view demonstrates the benefit of continuous process data because it shows not only that a trip occurred, but also how the process gradually moved toward the trip condition.

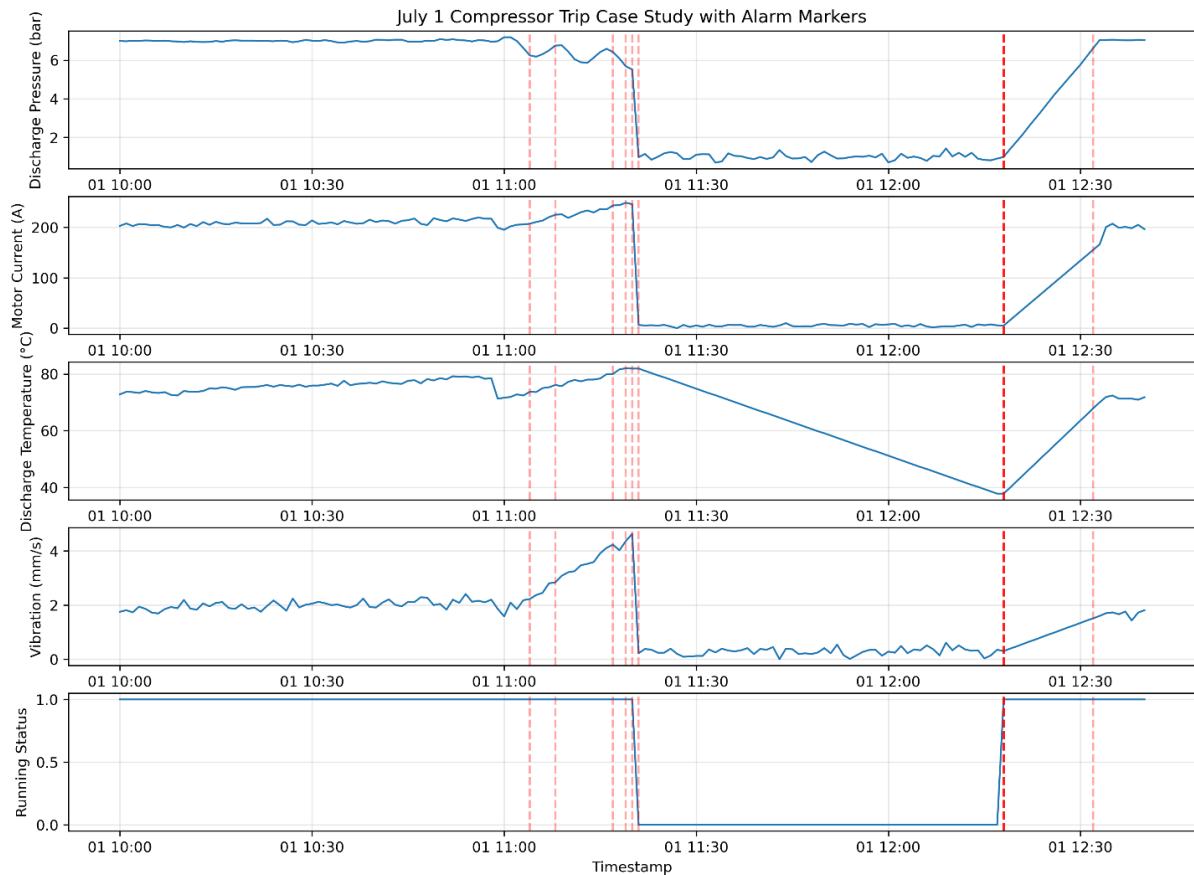


Figure 6: Focused July 1 Compressor Trip Window

8.3.3 Engineering Inference

The time-series analysis shows that continuous process data provides more information than event logs alone. An alarm log can indicate that a trip occurred, but trend data shows the progression before and after the event. By observing discharge pressure, flow, motor current, temperature, vibration, and running status together, it becomes possible to identify early warning patterns and understand the sequence of abnormal behavior.

This supports the need for storing continuous historian data for critical equipment. If similar data is available from actual plant operation, engineers can use it for abnormal trend detection, trip-sequence analysis, early warning systems, and preventive maintenance planning.

8.4 Signal Processing: Noise Filtering and Frequency Analysis

A basic signal filtering demonstration was performed to show how continuous sensor data can be processed to make underlying trends easier to observe. Industrial process signals often contain small point-to-point fluctuations due to sensor noise, process disturbances, control action, or mechanical variation. If such data is stored continuously, simple digital filtering methods can be applied to improve readability and support early identification of abnormal behaviour.

For this demonstration, discharge pressure and vibration were selected because pressure represents process delivery and vibration represents mechanical condition. A 15-minute moving average filter was applied to the July 1 trip window. Since the dataset was sampled at one-minute intervals, a 15-point moving average corresponds to a 15-minute smoothing window.

The filtered pressure signal reduced short-term fluctuations and made the overall pressure trend easier to observe. Similarly, the filtered vibration signal reduced point-to-point variation and made the gradual rise in vibration before the trip more visible. This demonstrates how basic signal processing can support condition monitoring by separating short-term noise from the underlying process trend.

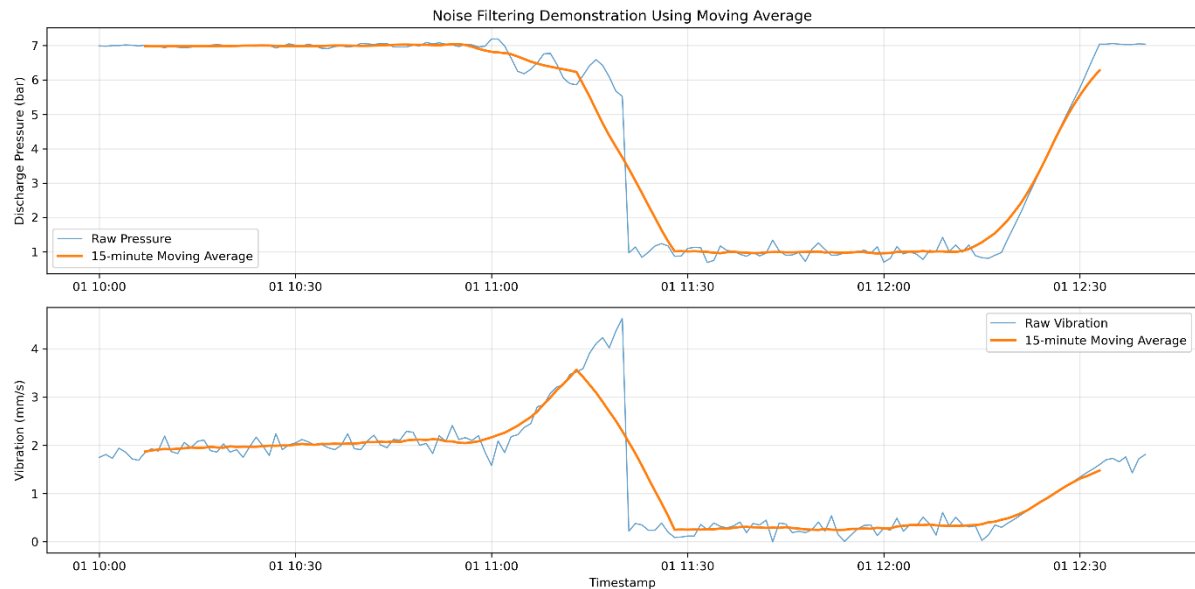


Figure 7: Raw and Filtered Signals Using a 15-minute Window

The purpose of this filtering step was not to replace the raw signal, but to provide an additional processed view that can help engineers interpret process behaviour more clearly. In practical plant monitoring, both raw and filtered signals can be useful: the raw signal preserves the original measurement, while the filtered signal helps reveal slower changes and developing abnormal trends.

To demonstrate the use of frequency-domain analysis on process data, Fast Fourier Transform (FFT) was applied to the pressure oscillation warning window. This window was selected because the alarm log contained a pressure oscillation warning related to discharge pressure and valve opening. Therefore, discharge pressure and valve opening percentage were chosen for FFT analysis.

FFT converts a signal from the time domain into the frequency domain, allowing repeated oscillatory components to be identified. Before applying FFT, the mean value of each signal was removed so that the analysis focused on variation around the operating point instead of the steady-state value. A Hanning window was also applied to reduce spectral leakage.

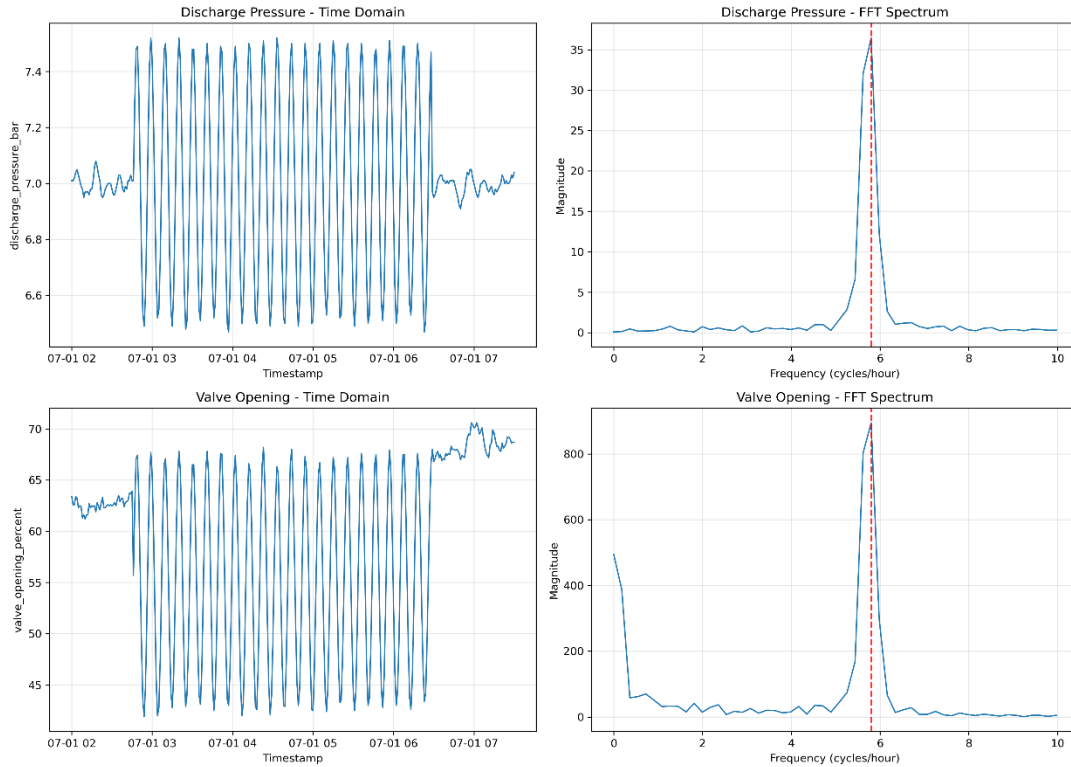


Figure 8: FFT-Based Oscillation Analysis

The time-domain plots show that both discharge pressure and valve opening repeatedly rise and fall during the selected window. The FFT spectra confirm this behaviour by showing a clear dominant frequency peak for both signals. The dominant frequency obtained for discharge pressure was approximately 5.625 cycles/hour, corresponding to an oscillation period of about 10.67 minutes. The valve opening signal also showed the same dominant frequency of approximately 5.625 cycles/hour, again corresponding to a period of about 10.67 minutes.

The purpose of applying FFT was not to perform detailed frequency-domain modelling, but to check whether the observed oscillation had a repeated dominant frequency. The presence of the same dominant frequency in both discharge pressure and valve opening indicates that the two signals are oscillating together. In a practical plant situation, such behaviour could be useful for identifying pressure hunting, control-loop oscillation, valve-control interaction, or repeated compressor loading/unloading patterns. This

demonstrates that storing continuous process data can allow engineers to move beyond simply observing that an alarm occurred and instead characterize the behaviour that caused the alarm.

8.5 Machine Learning-based Trip-Risk Prediction

A machine learning-based trip-risk prediction model was developed as part of the proof-of-concept analysis. The objective was to demonstrate how continuously stored compressor data could be used to estimate whether a compressor is likely to enter a trip-risk condition within the next 60 minutes. The target variable used for this task was `trip_risk_next_60min`, where 0 represented no expected trip-risk condition and 1 represented a trip-risk condition within the next 60 minutes.

Only process and sensor variables were used as model inputs. These included discharge pressure, air flow, discharge temperature, motor current, oil temperature, cooling water outlet temperature, valve opening percentage, load percentage, and vibration. The `health_state` column was deliberately excluded from the input features because it directly describes the operating condition of the compressor and could cause data leakage. The aim was to allow the model to learn from measured process behaviour rather than from a label that already contains condition information.

The dataset was split using a time-based method rather than random shuffling. The earlier portion of the data was used for training and the later portion was used for testing. This was done to better represent a practical plant scenario, where historical data would be used to train a model and future operating data would be used to test its prediction capability. Two models were trained: Logistic Regression and Random Forest Classifier. Logistic Regression was used as a simple baseline model, while Random Forest was used because it can capture nonlinear relationships between process variables.

The results show that the two models behaved differently. Logistic Regression achieved a high recall of 0.9667 for the trip-risk class, meaning it correctly identified most of the actual trip-risk samples. However, its precision was lower at 0.3867, which indicates that it also produced several false trip-risk warnings.

Table 8.4: ML Model Performance for Trip-Risk Prediction

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.9780	0.3867	0.9667	0.5524
Random Forest	0.9944	0.9091	0.6667	0.7692

The Random Forest model achieved a much higher precision of 0.9091 for the trip-risk class, meaning that when it predicted a trip-risk condition, the prediction was usually correct. However, its recall was lower at 0.6667, meaning that it missed more actual trip-risk samples compared to Logistic Regression.

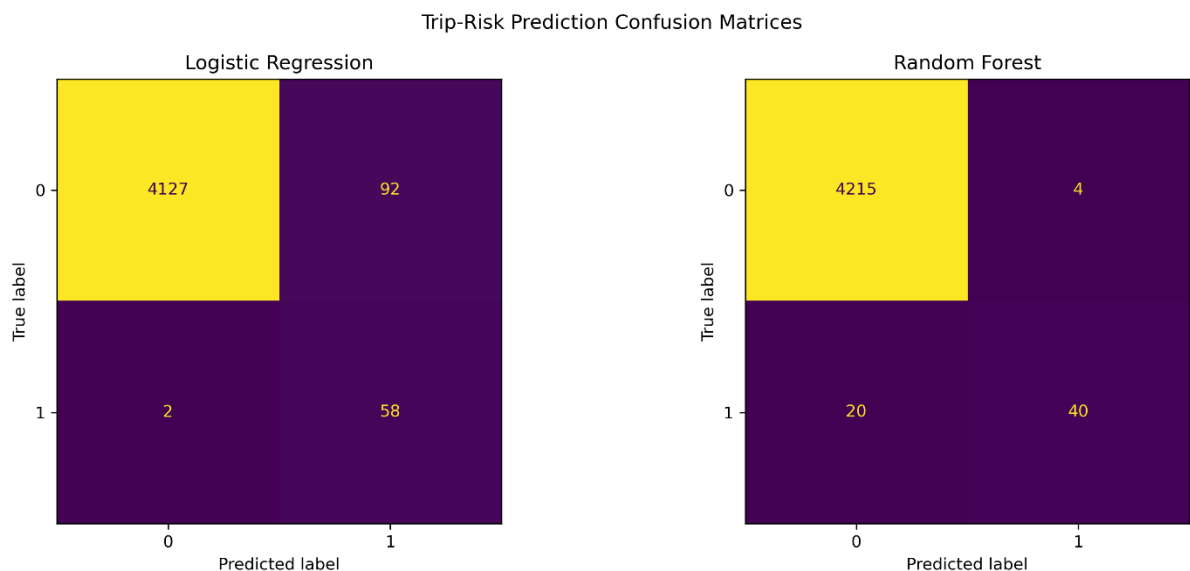


Figure 9: Confusion Matrices for the Prediction Models

The confusion matrices provide a clearer understanding of this trade-off. Logistic Regression correctly detected 58 out of 60 trip-risk samples but produced 92 false trip-risk alarms. Random Forest produced only 4 false trip-risk alarms but detected only 40 out of 60 trip-risk samples. Therefore,

Logistic Regression behaved more like a sensitive early-warning model, while Random Forest behaved like a more selective and high-confidence model.

In a practical plant-monitoring application, the choice between these two approaches would depend on the purpose of the system. If the priority is to avoid missing an upcoming trip, then a high-recall model such as Logistic Regression may be preferred. If the priority is to reduce nuisance alarms and generate only high-confidence warnings, then Random Forest may be more suitable. For compressor trip-risk monitoring, recall is especially important because missing an actual trip-risk condition may be more serious than generating an occasional early warning.

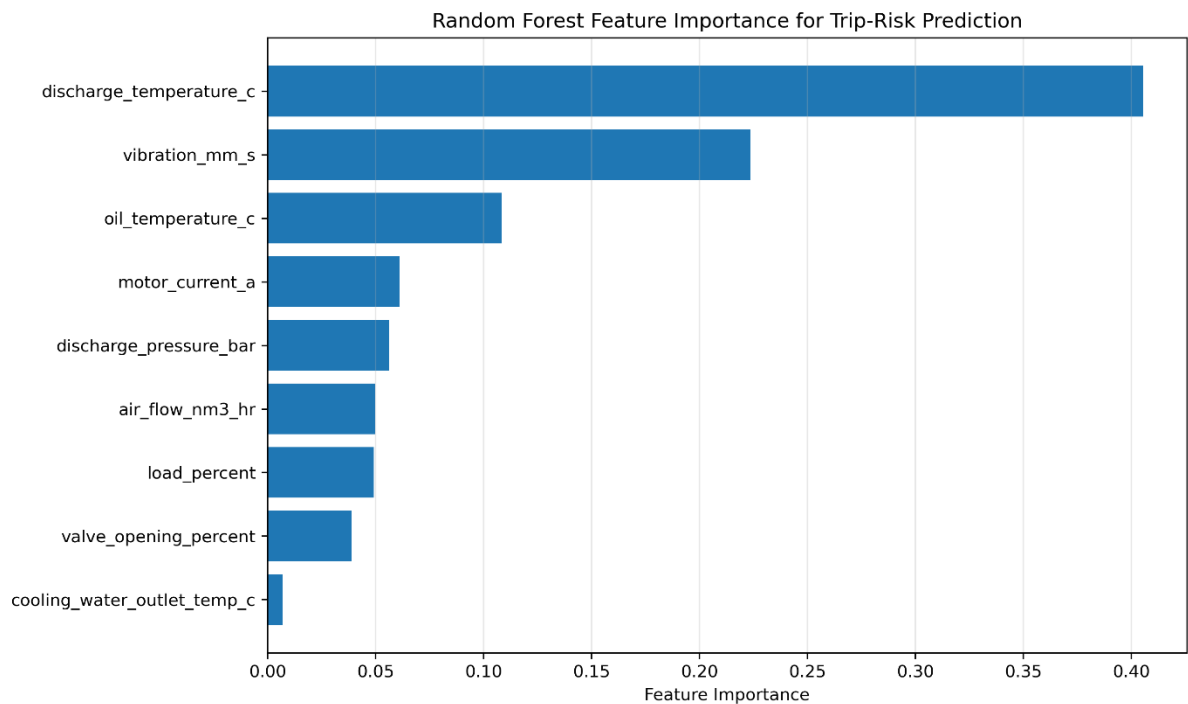


Figure 10: Random Forest Feature Importance for Trip-Risk Prediction.

The Random Forest feature importance plot showed that discharge temperature, vibration, oil temperature, motor current, and discharge pressure were among the most important variables for predicting trip-risk. This is consistent with the earlier time-series trend analysis, where Trip_Risk periods showed higher motor current, higher discharge temperature, higher oil

temperature, and higher vibration compared to Normal operating periods. For example, vibration increased from 1.60 mm/s in Normal condition to 2.83 mm/s during Trip_Risk periods, while motor current increased from 189.00 A to 212.75 A.

This result is meaningful from an engineering point of view because the model did not simply produce a prediction; it also highlighted which process variables were most relevant to the predicted trip-risk condition. In a real plant environment, these variables could be prioritized for continuous historian storage, dashboard monitoring, and early-warning rule development.

Overall, the ML analysis demonstrated that continuously recorded compressor data can support predictive monitoring. The models were not intended to represent a final deployable plant system, as the analysis was performed on synthetic data. However, the proof-of-concept showed that if similar data were stored from actual plant operation, machine learning could be used to identify developing trip-risk patterns and support preventive maintenance decisions.

8.6 Engineering Interpretation and Plant-Level Relevance

The proof-of-concept analysis demonstrated that structured continuous data can provide insights that are not available from alarm and event logs alone. While alarm logs indicate that an abnormal event occurred, time-series data shows how the system behaved before, during, and after the event. In the synthetic CAS compressor dataset, parameters such as discharge pressure, air flow, motor current, discharge temperature, oil temperature, vibration, valve opening, load, and running status helped identify abnormal behavior before trip conditions developed.

The trend analysis showed that trip-risk periods were associated with measurable changes in compressor behavior, such as reduced discharge pressure, increased motor current, higher discharge temperature, and increased vibration. This indicates that continuous historian data can help identify early warning patterns before a trip occurs. In an actual plant environment, such trends could support trip-sequence analysis, condition monitoring, and preventive maintenance planning.

The signal processing section further showed that basic techniques such as moving average filtering and FFT analysis can provide useful engineering insight. Filtering helped reveal slow changes in pressure and vibration by reducing short-term fluctuations. FFT analysis helped identify repeated oscillatory behavior between discharge pressure and valve opening, which could indicate pressure hunting, valve-control interaction, compressor loading/unloading patterns, or possible control-loop instability. Such analysis may be useful for control engineers while investigating repeated process disturbances or tuning control loops.

The machine learning section demonstrated how stored process data can be used for trip-risk prediction. Logistic Regression acted as a sensitive early-warning model, while Random Forest provided fewer false alarms and more selective predictions. In plant applications, the choice of model would depend on operational priority. A high-recall model may be useful when missing an upcoming trip is unacceptable, while a high-precision model may be preferred when the goal is to reduce nuisance warnings. Such models should not replace DCS protection logic or operator judgement, but could act as an additional advisory layer for predictive monitoring.

Although the signal processing and machine learning methods used in this proof-of-concept were intentionally simple, they demonstrate the potential for deeper analytical models in a large industrial plant. With sufficient real historian data, more advanced models could be used to identify hidden operating patterns, support predictive maintenance, improve reliability, and contribute to more efficient plant operation.

It is important to note that this analysis was performed using synthetic data and therefore does not represent the actual operating condition of the plant. The purpose of the project was to demonstrate the methodology and potential value of structured data storage. Actual deployment would require real plant historian data, validation by plant engineers, proper model testing, and integration with existing maintenance and operational practices.

Overall, the proof-of-concept supports the idea that continuous process data, when stored systematically, can help shift plant monitoring from reactive troubleshooting toward earlier detection, better maintenance planning, and more data-driven engineering decisions.

9. Technical Work Summary

The internship work was mainly focused on understanding the application of Electronics and Communication Engineering concepts in industrial automation, utility plant operation, control systems, and process data analysis. The work carried out involved control-room observation, field familiarization, documentation of process and control data, study of automation systems, and a proof-of-concept data-driven analysis project based on structured industrial data.

9.1 Summary of Technical Activities

Technical Activity	System / Area	Relevance
Observed DCS/SCADA screens, process graphics, trends, alarms, and valve positions	WTP, ASU, CAS	Understood real-time monitoring and operator interaction with plant systems
Studied signal flow from field instruments to DCS and actuators	Utility systems	Related 4–20 mA signals, Remote I/O, ADC/DAC conversion, fiber communication, and industrial networks to practical automation
Logged approximately 50 PID loops with tags, PV, MV, and remarks	ASU	Understood closed-loop control, process variables, manipulated variables, and control structure
Observed cascade control, differential pressure control, and control tuning activities	ASU	Studied advanced control behavior and practical PID-based troubleshooting
Studied trend graphs, alarm logs, and alarm setpoints	WTP and ASU	Understood how process data supports monitoring, fault detection, and operational decision-making
Studied historian/data availability and trip-related records	CAS	Understood limitations of event-based data and the need for continuous time-series storage
Developed a proof-of-concept data analysis project using synthetic industrial data	CAS-based compressor model	Demonstrated trend analysis, signal filtering, FFT-based oscillation detection, and ML-based trip-risk prediction

Rather than being limited to process familiarization, the internship involved studying how industrial automation systems operate from both a control-room and data-analysis perspective. The work connected field instruments, signal transmission systems, DCS/SCADA interfaces, control logic, process trends, alarm logs, and analytical methods into a single understanding of plant monitoring and control.

9.2 Technical Relevance

The technical work performed during the internship helped connect theoretical ECE concepts with real industrial applications. Concepts such as signal acquisition, industrial communication, real-time monitoring, feedback control, PID loop behavior, process data management, signal processing, and machine learning-based predictive monitoring were studied in the context of practical plant operation.

The most important learning was that industrial automation is not based on isolated instruments or software screens alone. It depends on the complete interaction between field devices, signal transmission systems, DCS/SCADA interfaces, control logic, alarms, trends, historian data, and operator actions.

The proof-of-concept analysis further demonstrated how structured data storage can extend the value of industrial automation systems. By applying trend analysis, moving average filtering, FFT-based oscillation detection, and basic machine learning models on synthetic compressor data, the project showed how continuous process data can support early fault identification, trip-risk monitoring, and future predictive maintenance applications.

10. Conclusion

The internship at Adani Kutch Copper Limited, Mundra, provided valuable practical exposure to the application of Electronics and Communication Engineering concepts in a large-scale industrial environment. The internship was carried out in the Control and Instrumentation department, with focus on utility systems such as the Water Treatment Plant, Air Separation Unit, and Compressed Air System. These systems helped in understanding how industrial processes are monitored, controlled, and maintained through automation, instrumentation, industrial communication networks, and control systems.

A major learning outcome of the internship was understanding the complete flow of information from field instruments to the control room. Field devices such as pressure, flow, level, and temperature instruments generate signals that are transmitted through Remote I/O panels, communication networks, and DCS/SCADA interfaces for monitoring and control. The study of 4–20 mA signal transmission, ADC/DAC conversion, fiber optic communication, and DCS-based visualization helped connect classroom concepts with real industrial applications.

The internship also provided practical exposure to closed-loop control systems. Through interaction with the DCS interface, multiple PID loops in the Air Separation Unit were studied, including pressure, flow, level, cascade, and differential pressure control loops. Special observations such as cascade control, differential pressure control, and trend-based troubleshooting during argon production helped in understanding how control systems are monitored and tuned in real plant conditions.

Another important learning was the role of process data in plant operation. Trend graphs, alarm logs, setpoints, and historian data were studied across different utility systems. This highlighted the importance of structured data storage for troubleshooting, reliability analysis, and predictive maintenance applications. The observations also showed that alarm logs alone provide limited information, whereas continuous time-series data can reveal how process variables behave before, during, and after abnormal events.

As an extension of this learning, a proof-of-concept data analysis project was carried out using a synthetic compressed air system dataset. The project demonstrated how structured time-series data, alarm/event logs, and calibration records could be used for trend analysis, signal filtering, FFT-based oscillation detection, machine-learning-based trip-risk prediction, and instrument drift monitoring. Although the dataset was synthetic and did not represent actual plant operating data, the analysis showed the potential value of storing plant data in a structured and analysis-ready format.

The proof-of-concept also helped connect signal processing and machine learning concepts with industrial applications. Moving average filtering helped separate short-term fluctuations from underlying process trends, while FFT analysis demonstrated how oscillatory behavior in process and control signals can be identified. Machine learning models such as Logistic Regression and Random Forest showed how stored process data could be used to estimate trip-risk conditions and identify important variables contributing to abnormal behavior.

Overall, the internship helped in developing a practical understanding of industrial automation beyond theoretical study. It demonstrated how sensors, transmitters, communication systems, controllers, actuators, PID loops, alarm systems, historian data, and analytical tools work together to support safe and efficient plant operation. The experience strengthened the understanding of Electronics and Communication Engineering in the context of process industries and provided a strong foundation for further work in industrial automation, control systems, signal processing, and data-driven plant monitoring.